

Generation Interconnection Request System Impact Study Report

For Queue Project Q15-0547

O'Fallon Renewable Energy Center 138 kV
November 03, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0547. The Interconnection Customer proposes a 184.8 MW wind facility interconnecting at the O'Fallon Renewable Energy Center 138 kV substation in the Pennsylvania area. The request was received on December 04, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 304 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,307,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Curecanti / Eliza Creek 345.0 kV Line	Tombstone / Bixby Hill 500.0 kV Line	145.8	1,933	6.4
Elk River / Merury 345.0 kV	Bixby Hill / Tombstone 500.0 kV	131.9	1,196	10.6
Curecanti / Moser Generating Station 345.0 kV	Tombstone / Bixby Hill 500.0 kV L	123.5	2,357	6.1
Eliza Creek / Ontelaunee Energy Ctr 345.0 kV	Tombstone / Bixby Hill 500.0 kV	111.7	2,310	5.1
Tombstone / Marcus Hook Energy Lp 500.0 kV	Moser Generating Station / Curecanti 345.0 kV Line	102.9	3,797	8.7
Marcus Hook Energy Lp / Guilbeau Electric Substation 500.0 kV	Irvine / Marcus Hook Energy Lp 500.0 kV	101.8	2,785	5.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Curecanti / Eliza Creek 345.0 kV Line	Rebuild line with higher-capacity conductor	\$27,419,000	3% (\$737,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Elk Riv Substation / Mercury 345.0 kV	Rebuild line with higher-capacity conductor	\$20,127,000	6% (\$1,289,000)
Curecanti / Moser Generating Station 345.0 kV	Rebuild line with higher-capacity conductor	\$35,407,000	2% (\$880,000)
Eliza Creek / Ontelaunee Energy Ctr 345.0 kV	Rebuild line with higher-capacity conductor	\$19,372,000	3% (\$509,000)
Tombstone / Marcus Hook Energy Lp 500.0 kV	Reconductor line	\$42,948,000	1% (\$595,000)
Guilbeau Electric / Marcus Hook Energy Lp 500.0 kV Line	Reconductor line	\$14,747,000	2% (\$297,000)
	Total allocated to project		\$4,307,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.